Scrambling, not athermality, protects quantum information under generic measurement: chaotic states outperform quantum many-body scar codes

Hikaru Wakaura and Taiki Tanimae

QIRI (Quantum Integrated Research Institute Inc.), Tokyo 107-0061, Japan

Monitored quantum many-body systems protect quantum information from projective measurements exactly when their unitary dynamics scramble it, placing the measurement-induced transition on the same footing as quantum error correction. Because the eigenstate thermalization hypothesis endows chaotic eigenstates with approximate Knill–Laflamme structure, it is natural to ask whether quantum many-body scars—non-thermal eigenstates that stay weakly entangled and dynamically coherent—form especially good codes. Whether they protect information better or worse than the surrounding thermal states has not been settled quantitatively. Here we show that scrambling, not athermality, is what protects an encoded qubit: under monitored dynamics, the reference-qubit coherent information of a logical qubit encoded in the PXP scar subspace lies below that of an energy-matched thermal encoding for every measurement we test. The deficit is mild for local measurement, largest for collective measurements aligned with the emergent $\mathrm{su}(2)$ order parameter that couples the code states, and—decisively—survives even a fine-tuned Casimir measurement for which the scar code is a nominal decoherence-free subspace, because approximate scars carry irreducible Casimir variance. An exactly solvable spin-1 model isolates that variance as the control parameter: only zero-variance *exact* scars are protected, and even they gain no systematic advantage under generic local measurement. The eigenstate-thermalization-to-error-correction correspondence thus holds directly in the monitored-circuit setting, and scarring does not aid protection against generic measurement. Scrambling is therefore the design resource for measurement-robust encoding, and weak-ergodicity-breaking subspaces are liabilities rather than assets for measurement-based quantum memories.

Hikaru Wakaura: h.wakaura@qiri.co.jp
Taiki Tanimae: t.tanimae@qiri.co.jp

1 Introduction

The competition between scrambling unitary dynamics and local measurements drives a sharp transition in how much quantum information a many-body system retains [1–7]. This measurement-induced phase transition (MIPT) is now understood as a coding transition: below a critical measurement rate the dynamics act as an emergent quantum error-correcting code that hides logical information non-locally, protecting it from measurements that act as erasure errors [8–10]. The order parameter is information-theoretic—the coherent information, or equivalently the entanglement of a reference qubit maximally entangled with the encoded information [9, 11, 12].

The coding power of chaotic dynamics is inherited by chaotic *eigenstates*. The eigenstate thermalization hypothesis (ETH) forces local operators to have smooth diagonal and exponentially small off-diagonal matrix elements [13–16], which is exactly the approximate Knill–Laflamme condition [17]; thermal eigenstates therefore form approximate codes whose error scales with the system's chaoticity [18–20]. Quantum many-body scars are the conspicuous exception to ETH: a measure-zero tower of atypically weakly entangled, dynamically reviving eigenstates embedded in an otherwise thermal spectrum [21–25]. Scars can be embedded in decoherence-free subspaces [26–28], and projective measurement can even enhance their revivals [29], suggesting they might be privileged carriers of quantum information under monitoring.

However, whether a logical qubit stored in the scar subspace is protected better or worse than one stored in the surrounding thermal bulk—at matched energy, in the same monitored dynamics, quantified as a channel capacity—has not been determined. In this study, we compute the reference-qubit coherent information $\mathcal{C}_R$ of scar and thermal codes of the monitored PXP chain across local, collective, and Casimir measurement models. We find that thermal codes protect information better than scar codes in every case; that an apparent scar advantage seen against a single thermal reference is an artifact of an atypical baseline and vanishes under ensemble averaging; and that the scar's emergent-su(2) structure is a liability, not a shield, because its generator couples the code states and the scars' Casimir variance defeats even a fine-tuned decoherence-free encoding.

The result establishes the ETH-to-error-correction correspondence as an operational statement about monitored circuits and removes weak-ergodicity-breaking subspaces from the menu of candidate measurement-robust quantum memories.

2 Model and diagnostics

2.1 Model, encoding, and monitored dynamics

We study the PXP chain of L sites in the Rydberg-blockaded (constrained) Hilbert space of dimension $F(L{+}2)$ (Fibonacci), with

$$H = \sum_i P_{i-1}\, X_i\, P_{i+1}, \qquad P = |0\rangle\langle 0|, \qquad (1)$$

whose $\mathbb{Z}_2$ Néel state exhibits the canonical scar revivals [21, 22, 30, 31] (period $t \simeq 4.7$, fidelity $\simeq 0.74$ at $L = 12$). We identify the $L{+}1$ scar-tower eigenstates by their anomalous overlap with $|\mathbb{Z}_2\rangle$.

A logical qubit is encoded in a two-dimensional subspace $\{|0_L\rangle, |1_L\rangle\}$ and maximally entangled with a reference R, $|\Psi_0\rangle = (|0\rangle_R |0_L\rangle + |1\rangle_R |1_L\rangle)/\sqrt{2}$. The system evolves by alternating steps of PXP evolution $e^{-iH\,dt}$ and random projective measurement; per quantum trajectory the joint state stays pure and the reference entanglement entropy $S_R = -\mathrm{Tr}\,\rho_R \log_2 \rho_R \in [0, 1]$ measures the surviving logical information. Its trajectory average

$$\mathcal{C}_R(p) = \langle S_R \rangle_{\mathrm{traj}} \qquad (2)$$

is the single-qubit coherent information of the monitored channel: $\mathcal{C}_R \to 1$ recoverable, $\mathcal{C}_R \to 0$ lost. We take $dt = 0.6$ and measurement probability p per site (local) or per operator (collective) per step. To control the energy-mismatch confound, thermal codes are drawn from generic eigenstates energy-matched to the two scar-rung energies within a tolerance ± 0.5 (realized offsets at $L{=}14$: 0.22 on average, 0.47 at most); **the thermal baseline is averaged over an ensemble of** $\sim$ 10–20 **such pairs**, without which a single atypical pair inverts the conclusion (Sec. 3). The comparison is insensitive to this tolerance: tightening the window to ± 0.1 leaves the scar deficit unchanged, and within the ensemble $\mathcal{C}_R$ is uncorrelated with the residual offset (Appendix C).

We study $L = 10$–18 (constrained dimension up to $F(20) = 6765$) with 40 evolution–measurement steps per trajectory and average $\mathcal{C}_R$ over 120–400 Born-sampled quantum trajectories. Reported statistical error bars are the standard error of the mean over trajectories; the thermal band is the standard deviation of $\mathcal{C}_R$ over the ensemble of energy-matched thermal pairs. All runs use fixed random seeds and are reproducible from the accompanying code.

Statically we probe the Knill–Laflamme violation of a code under an error set $\{E_a\}$ through the logical leak $\langle 0_L|E_a|1_L\rangle$ and the distinguishability $\langle 0_L|E_a|0_L\rangle - \langle 1_L|E_a|1_L\rangle$.

2.2 A model-independent criterion

A projective measurement of a Hermitian operator O leaves the encoded qubit literally untouched only if the code is an eigenspace of O with a single shared eigenvalue; more generally, the qubit survives a single shot if and only if the Knill–Laflamme conditions hold for the projector set $\{\Pi_g\}$ of O's eigenspaces [17]. A measurement *resolves*—and hence degrades—the code through two channels: *(i) eigenvalue distinguishability*, when the two logical states' outcome distributions $P_a(g) = \langle a_L|\Pi_g|a_L\rangle$ differ; and *(ii) non-invariance*, when a sector-restricted overlap $\langle 0_L|\Pi_g|1_L\rangle$ is nonzero, coherently mixing the logical states. Equal means and a vanishing global leak $\langle 0_L|O|1_L\rangle$ are therefore insufficient: a code spread over many O-sectors—a large O-variance—is collapsed by the shot, and survival then requires the sector-wise conditions in *every* occupied sector, which is why a true decoherence-free subspace demands logical states that are O-eigenstates sharing one eigenvalue. This criterion is independent of the microscopic model and organizes the results below: thermal states, spread broadly and near-identically in a local basis, are close to optimal codes, while scars are penalized whenever the measured operator resolves their atypical structure. Both channels combine into a single exact, simulation-free scalar—the reference entropy $s_1(O)$ surviving one O-measurement, which equals unity precisely when the sector-wise conditions hold—and s_1 predicts the monitored $\mathcal{C}_R$ across resolving measurements (Appendix H).

3 Results

3.1 Thermal codes win: the apparent scar advantage is a baseline artifact

Figure 1 compares $\mathcal{C}_R(p)$ for the scar code against the thermal ensemble. For both local-Z and collective (staggered-Z) monitoring the scar curve lies *below* the thermal-ensemble band at all p. A single nearest-in-energy thermal pair sits at the 0th percentile of the ensemble ($\mathcal{C}_R = 0.53$ versus ensemble mean 0.87 at L=14, p=0.10, collective), which is why an unaveraged comparison spuriously favors the scar.

3.2 The disadvantage persists to the largest accessible size

Figure 2 shows $\Delta\mathcal{C}_R \equiv \mathcal{C}_R^{\mathrm{scar}} - \langle\mathcal{C}_R^{\mathrm{therm}}\rangle$ at $p = 0.10$. It is negative for all $L = 10$–18 and both channels: -0.008 to -0.06 (local) and -0.25 to -0.32 (collective). The thermal-ensemble band excludes zero at every size for the collective channel, and at every size except $L = 12$ for the local one (there -0.008 ± 0.009, consistent with zero). At the largest size $L = 18$ (constrained dimension 6765) the gap is -0.05 (local) and -0.30 (collective), the collective gap essentially size-independent: the refutation is not a finite-size artifact.

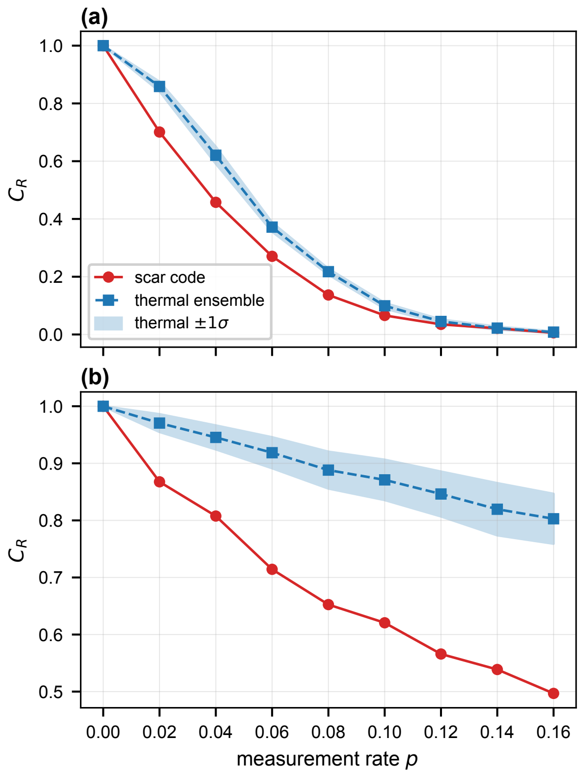

Figure 1: **Thermal codes protect information better than scar codes.** Reference-qubit coherent information $\mathcal{C}_R$ versus measurement rate p for the scar code (red) and the thermal ensemble (blue; band is $\pm1\sigma$ over the ensemble) under (a) local-Z and (b) collective staggered-Z monitoring at $L = 14$. The scar lies below the thermal band throughout.

3.3 Even the maximal scar code loses

We emphasize that this single-qubit comparison is the most favorable one for the scar: the scar tower is only $(L{+}1)$-dimensional, so it can host at most $\log_2(L{+}1)$ logical qubits—a sub-extensive capacity, against the exponentially large thermal bulk. A scar subspace therefore cannot support an extensive code even before dynamics; the per-qubit comparison already settled here is its best case, and it loses.

To confirm this directly, we encode the *maximal* scar code—all $k = \lfloor\log_2(L{+}1)\rfloor$ logical qubits, i.e. 2^k tower states maximally entangled with a k-qubit reference—and compare its protected-information density S_R/k against a same-dimension thermal code under local monitoring (Fig. 3). The scar loses at every rate: at $L = 14$ ($k = 3$), $\Delta(S_R/k) = -0.17, -0.14, -0.06$ at $p = 0.02, 0.04, 0.08$, with the thermal ensemble tightly determined ($\sigma \lesssim 0.01$), and the gap

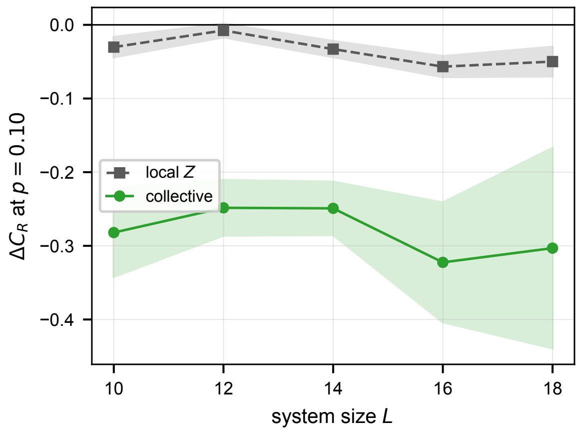

Figure 2: **The scar disadvantage $\Delta\mathcal{C}_R < 0$ at every size.** Scar-minus-thermal coherent information at $p = 0.10$ versus L for local (grey) and collective (green) monitoring; shading is the thermal-ensemble spread.

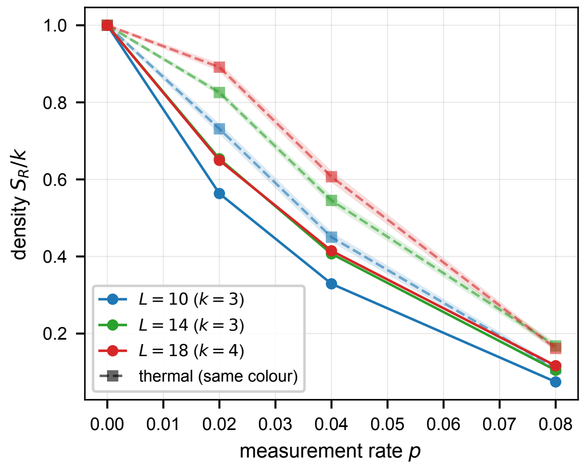

Figure 3: **The maximal scar code loses even as an extensive memory.** Protected-information density S_R/k for the maximal $k = \lfloor \log_2(L+1) \rfloor$-qubit scar code (solid) and a same-dimension thermal code (dashed, $\pm 1\sigma$ band) versus p under local monitoring, for $L = 10, 14, 18$. Scar lies below thermal throughout.

persists across $L = 10, 14, 18$. The gap is even larger under collective staggered-Z monitoring ($\Delta(S_R/k) = -0.32$ to -0.43 at $L = 14$), while under the fine-tuned Casimir J^2 the two codes are comparable ($|\Delta| \lesssim 0.03$; Appendix I)—the same exception seen in the single-qubit and spin-1 analyses. Even the largest code the scar subspace admits is thus a worse monitored memory than a generic thermal code of equal size under any generic measurement.

3.4 Mechanism: both channels resolve the scar code

The PXP scar realizes both channels of the criterion while thermal codes realize neither. Channel (ii) dominates under the collective staggered operator $M = \sum_i (-1)^i Z_i$—the staggered magnetization, order parameter of the emergent su(2) scar dynamics [32]—which carries a large logical leak $|\langle 0_L | M | 1_L \rangle| = 1.73$ between the scar rungs (versus ≈ 0 for thermal codes) and coherently mixes them. This leak is a one-sided suppressor of $\mathcal{C}_R$ [Fig. 4(b)]: every high-leak code is poorly protected, whereas low leak permits any value. Channel (i)—eigenvalue distinguishability through O-variance—is isolated by the Casimir measurement below. Accompanying both is a weak monotone trend [Fig. 4(a)]: protection rises with the code's spread in the measurement basis, as ETH predicts, so that unstructured thermal codes are close to optimal.

The full (p, ℓ) dependence, with ℓ the measurement support size (block-sum of ℓ sites; ℓ=1 local, ℓ=L global), is negative throughout: the scar never wins (Fig. 7, Appendix E).

3.5 The decoherence-free-subspace rescue fails

A true DFS [26, 27] for the scar code would restore an advantage. The emergent Casimir $J^2 = \frac{1}{2}(H^+H^- + H^-H^+) + J_z^2$, built from the exact split $H = H^+ + H^-$ [32], gives the chiral-symmetric scar pair $(|{+E}\rangle, |{-E}\rangle)$ equal expectation and vanishing leak ($\Delta\langle J^2\rangle = 0.000$, leak $= 0.052$)—a nominal DFS. Yet under J^2 monitoring the scar code still loses, $\Delta\mathcal{C}_R = -0.18, -0.13, -0.09$ at $p = 0.04, 0.08, 0.12$. The reason is that equal expectation and zero leak are necessary but not sufficient: a genuine DFS requires the code states to be *eigenstates* of the measured operator. Approximate PXP scars carry irreducible Casimir variance (Var $J^2 = 32.5$, against an ensemble mean of 6.1, range 2–17, for the energy-matched thermal states), so J^2 measurement collapses them more strongly than it does thermal states. No Casimir measurement can be a true DFS for approximate scars.

3.6 A second model isolates the variance as the culprit

To test generality and to isolate this mechanism we repeat the analysis in the spin-1 Schecter–

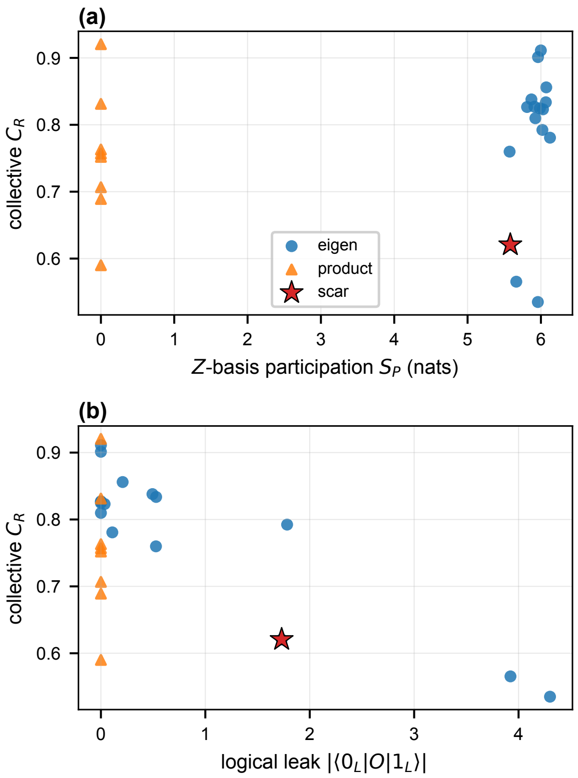

Figure 4: **What controls collective protection.** Collective $\mathcal{C}_R$ versus (a) Z-basis participation and (b) logical leak $|\langle 0|O|1\rangle|$ over an eigenstate sweep (blue), product states (orange), and the canonical scar (star). Higher participation raises $\mathcal{C}_R$; large leak suppresses it. The scar is a low-$\mathcal{C}_R$ outlier on both axes.

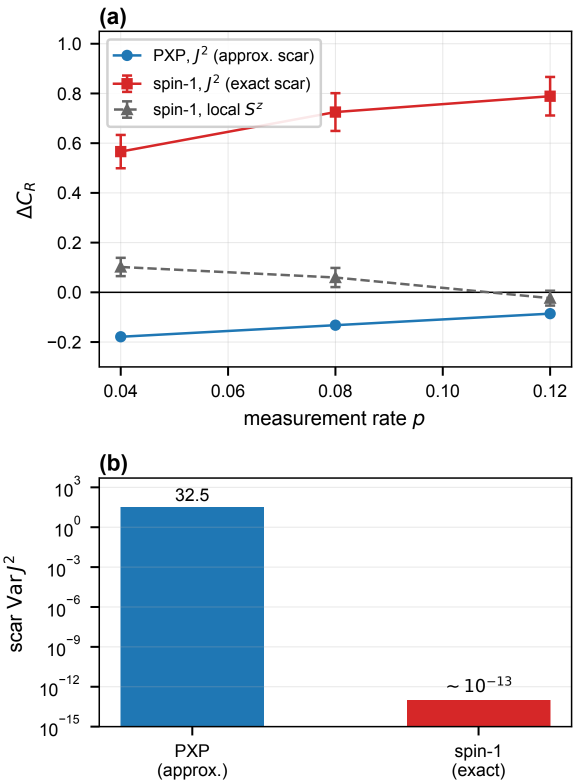

Figure 5: **Exact-su(2) scars evade the Casimir-variance obstruction.** Second model (spin-1 Schecter–Iadecola, $L = 6$, 20-pair thermal ensemble). (a) Scar–thermal coherent information $\Delta\mathcal{C}_R$ versus p (error bars are the ensemble SEM). Under J^2 monitoring PXP scars lose (blue) while exact spin-1 scars win perfectly (red); under generic local S^z the exact-scar and thermal codes are comparable ($|\Delta\mathcal{C}_R| \lesssim 0.1$, grey). (b) Scar Casimir variance, $\simeq 32$ (PXP) versus $\sim 10^{-13}$ (exact) on a log scale—the control parameter of the rescue.

Iadecola magnet [33], whose scar tower $|S_n\rangle = (Q^+)^n |\Omega\rangle$, $Q^+ = \sum_j (-1)^j (S_j^+)^2$, is an *exact* su(2) multiplet. Two facts then hold to machine precision: the scars are exact eigenstates, and the Casimir J^2 is exactly constant on the tower with *zero* variance on every scar—against $\mathrm{Var}\, J^2 \simeq 32$ for PXP [Fig. 5(b)]. This flips the Casimir rescue [Fig. 5(a)]: the exact-su(2) scar code is now a genuine decoherence-free subspace—its two logical states are identical J^2 eigenstates—and is *perfectly* protected under J^2 monitoring, $\mathcal{C}_R^{\mathrm{scar}} = 1$ at every p, giving $\Delta\mathcal{C}_R = +0.6$ to $+0.9$ ($L = 6, 8$). The rescue that fails for approximate PXP scars succeeds exactly when the Casimir variance vanishes, confirming that this variance—not scarring itself—controls it. This is parameter-independent: for $(h, D) = (0.5, 0.3)$ and $(1.5, 0.0)$ the Casimir variance again vanishes identically and $\mathcal{C}_R^{\mathrm{scar}} = 1$ under J^2. The advantage is, however, confined to this fine-tuned Casimir probe: under generic local S^z monitoring the exact-scar code shows no *systematic* advantage—a small scar-favorable difference at the lowest rate ($\Delta\mathcal{C}_R \simeq +0.10$ to $+0.17$ at $p = 0.04$, $L = 6, 8$) decays and changes sign as p grows—so scar structure again fails to protect against generic local measurement.

3.7 Calibration to prior monitored-scar work

Reproducing the monitored-Néel entanglement transition and mapping our per-step rate to continuous time via $\gamma = p/dt$ yields a critical rate $\gamma_c \simeq 0.083$ ($p_c \simeq 0.05$), the same order as the $\gamma_c \simeq 0.013 \pm 0.002$ reported for continuous-time projective σ^z monitoring of PXP scars [29]; the factor ~ 6 reflects protocol and estimator differences we have not converted quantitatively—stroboscopic evolution interleaved with per-site measurement versus continuous-time monitoring, and a finite-size crossing estimate versus a full finite-size-scaling analysis.

3.8 Experimental accessibility

The negative result is testable on the platform where PXP scars were first observed [21]. Programmable Rydberg-atom arrays natively realize the blockaded dynamics, local-Z monitoring is site-resolved projective readout interleaved at a tunable rate, and the sizes studied here ($L \leq 18$) are modest by current standards. Encoding in the scar subspace is accessible through the Néel revival manifold, and the retained coherence can be estimated by interferometric or randomized-measurement protocols. While preparing individual thermal eigenstates is impractical, thermal-like codes can be prepared operationally by pre-scrambling product states under the chaotic dynamics itself, and the scar code's excess fragility appears directly as a faster purification of the Néel-manifold encoding.

4 Conclusion

We quantified, as a monitored-channel coherent information, how well a logical qubit stored in the PXP quantum-many-body-scar subspace is protected against random measurement, benchmarked against an ensemble-averaged thermal code at matched energy—to our knowledge the first such head-to-head coding comparison of scar versus thermal subspaces under monitoring. Thermal codes protect information better than PXP scar codes at every size and under every measurement model, by $\Delta\mathcal{C}_R \simeq -0.05$ (local) to -0.3 (collective) at $p = 0.10$, up to $L = 18$; the scar's emergent-su(2) order parameter supplies a logical leak of 1.73 that makes it especially fragile, and even a fine-tuned Casimir DFS fails because approximate scars retain Casimir variance ~ 30. This realizes the eigenstate-thermalization-to-approximate-error-correction correspondence [18–20] as an operational statement about monitored circuits: chaos, not athermality, is the coding resource. Any program that would exploit weak ergodicity breaking for measurement-robust quantum memory should expect to lose to a generic chaotic subspace; the relevant resource is the code's spread in the measurement basis, and structured non-thermal towers whose generators connect the code states are actively harmful. A second, exactly-solvable spin-1 model [33] whose tower closes the algebra exactly confirms the obstruction is variance-controlled—there the Casimir measurement protects the scar perfectly, while under generic local measurement scars retain no advantage. Whether this fragility survives generic non-projective noise channels and finite-temperature encodings [25, 34] is the natural next question.

Acknowledgments

This research received no specific grant from any funding agency in the public, commercial, or not-for-profit sectors.

Author contributions

H.W. and T.T. conceived the study. H.W. developed the methodology, implemented the software, performed all computations and analysis, produced the figures, and wrote the manuscript.

Data and code availability

The simulation package, drivers, and the data underlying all figures are openly available at doi:10.5281/zenodo.21642056 (source: `https://github.com/deeptell-inc/chaosec`), enabling one-command reproduction of every result reported here.

A Constrained basis, PXP spectrum, and scar identification

The PXP chain is diagonalized in the Rydberg-blockaded basis of bit strings with no two adjacent excitations (dimension $F(L{+}2)$, Fibonacci; open boundaries). We benchmark the implementation against the canonical $\mathbb{Z}_2$-Néel revival: for $L = 12$ the return fidelity $|\langle\mathbb{Z}_2|e^{-iHt}|\mathbb{Z}_2\rangle|^2$ peaks at $t \simeq 4.70$ with height 0.743, reproducing the literature. The $L{+}1$ scar-tower eigenstates are selected as the states with anomalously large $|\langle\mathbb{Z}_2|E_k\rangle|^2$; the tower forms an approximately equally spaced ladder at $E \simeq$

$0, \pm 1.33, \pm 2.66, \pm 4.0$. PXP possesses an exponentially large $E = 0$ degeneracy ($8, 13, 21$ zero modes at $L = 10, 12, 14$); scar and thermal codes are therefore built from positive-energy rungs ($|E| > 0.8$) to avoid this manifold.

B Monitored-trajectory algorithm

A logical qubit maximally entangled with a reference R is stored as $\phi \in \mathbb{C}^{d \times 2}$, with column r holding $\langle r_R | \Psi \rangle$ (an unnormalized system vector of dimension d). Each step applies the dense propagator $U = e^{-iH\,dt}$ to both columns and then, with probability p, performs a projective measurement: for a diagonal operator with values $\{v\}$ we Born-sample an eigenvalue from weights $\sum_r \sum_{q:v_q=v} |\phi_{qr}|^2$ and project. The reference density matrix is $\rho_R = \phi^\dagger \phi$ (a 2×2 Hermitian matrix of unit trace), and $S_R = -\mathrm{Tr}\, \rho_R \log_2 \rho_R$. We report $\mathcal{C}_R = \langle S_R \rangle$ over 120–400 Born-sampled trajectories with fixed seeds; error bars are the standard error over trajectories. As a consistency check, $p = 0$ yields $\mathcal{C}_R = 1$ identically (a unitary on the system cannot reduce the reference entanglement), and $\mathcal{C}_R \to 0$ at large p.

For a general (non-diagonal) Hermitian measurement operator O (e.g. the su(2) Casimir J^2) we precompute its eigenbasis once, rotate ϕ into it, project onto a Born-sampled eigenvalue group, and rotate back.

C Thermal-ensemble baseline and the single-pair artifact

The thermal baseline must be averaged over many energy-matched thermal pairs. The naive choice—the single non-scar eigenstate nearest each scar-rung energy—is systematically biased toward atypical, near-scar states with anomalously low $\mathcal{C}_R$. At $L = 14$, $p = 0.10$, collective monitoring, that single pair gives $\mathcal{C}_R = 0.53$, which sits at the 0th percentile of a 20-pair ensemble whose mean is 0.87 ± 0.03 (minimum 0.81). Averaging over the ensemble reverses the sign of $\Delta\mathcal{C}_R$ and is essential to the conclusion; all results in the main text use the ensemble baseline.

C.1 Energy-matching tolerance

The ensemble is selected within a finite energy window, not at exactly the scar-rung energies: each member is a generic (non-tower, $|\langle \mathbb{Z}_2 | E \rangle|^2 < 0.02$) eigenstate within ± 0.5 of one of the two targets. At $L = 14$ the realized offsets are 0.22 on average and 0.47 at most, so the matching is approximate. Three checks, all at the headline point (L=14, p=0.10, 12 pairs, 250 trajectories, identical seeds), show this tolerance does not produce the scar deficit. *(i)* Tightening the window from ± 0.5 to ± 0.25 and ± 0.10 (mean offset $0.22 \to 0.13 \to 0.06$) leaves the deficit essentially unchanged: $\Delta\mathcal{C}_R = -0.033$, -0.034, -0.031 (local) and -0.25, -0.23, -0.21 (collective); at the tightest matching the gaps remain ≈ 2.4 and ≈ 2.8 ensemble standard deviations below zero. *(ii)* Within the ensemble the per-pair $\mathcal{C}_R$ is uncorrelated with the pair's mean absolute offset [Spearman $\rho = +0.05$ ($p = 0.88$, local) and 0.00 ($p = 1.00$, collective) at ± 0.5], so the thermal band reflects state-to-state typicality, not residual energy dependence. *(iii)* The weak drift with the window (collective thermal mean $0.87 \to 0.83$) slightly *shrinks* the gap, the opposite of an energy-mismatch artifact inflating it. The spin-1 and extensive-code analyses use analogous windows (± 0.8 and ± 0.5). The audit is reproduced by `scripts/energy_window_robustness.py`.

D Static Knill–Laflamme diagnostics

For a code $\{|0_L\rangle, |1_L\rangle\}$ and error set $\{E_a\}$ we quantify correctability through the logical leak $u_a = \langle 0_L | E_a | 1_L \rangle$ and distinguishability $d_a = \langle 0_L | E_a | 0_L \rangle - \langle 1_L | E_a | 1_L \rangle$. Under the local-$Z$ set, energy-matched scar codes have a larger root-mean-square violation than thermal codes (ratio 1.4–2.4 for $L = 10$–14), consistent with ETH-based approximate error correction. Under the collective staggered operator $M = \sum_i (-1)^i Z_i$ the scar rungs, being connected by the emergent su(2) generator, carry a large leak $|\langle 0_L | M | 1_L \rangle| = 1.73$ (versus ≈ 0 for thermal codes), which drives the strong scar penalty under SGA-aligned monitoring.

The two channels of the main-text criterion are diagnostically distinct. The diagonal outcome-distribution overlap alone—the Bhattacharyya coefficient $\mathrm{BC} = \sum_o \sqrt{P_0(o) P_1(o)}$ of the two logical states under the collective M—does *not* pre-

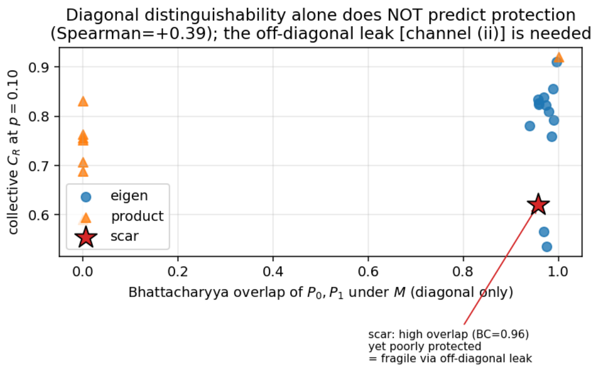

Figure 6: Collective $\mathcal{C}_R$ versus the diagonal Bhattacharyya overlap of the two logical states under M ($L = 14$). The scar (star) has near-unit overlap yet low $\mathcal{C}_R$: its fragility is off-diagonal [channel (ii)], invisible to BC.

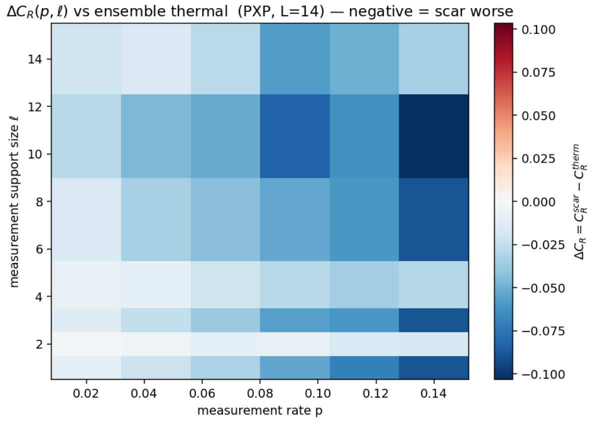

Figure 7: $\Delta\mathcal{C}_R(p, \ell)$ against the ensemble thermal baseline (PXP, $L = 14$). Blue is $\Delta\mathcal{C}_R < 0$ (scar worse) everywhere.

dict $\mathcal{C}_R$ (Fig. 6): the scar has a high overlap $\mathrm{BC} = 0.96$ yet is poorly protected, because its M-fragility comes from channel (ii), the off-diagonal leak, which BC cannot see. Both channels are therefore required; no single scalar (participation, variance, or leak alone) predicts protection across all codes.

E Measurement-support phase diagram

Figure 7 maps $\Delta\mathcal{C}_R(p, \ell)$ where ℓ is the support size of a block-sum measurement $\sum_{i \in W} Z_i$ over a random contiguous window W of ℓ sites (ℓ=1 local, ℓ=L global uniform), holding the per-step event count fixed. Against the ensemble thermal baseline it is negative throughout: the scar code never wins, and the disadvantage deepens with both p and ℓ. (The staggered SGA operator of the main text produces a larger penalty than the uniform-type block sum because the latter has vanishing logical leak on the scar rungs.)

F Calibration of the measurement rate

To connect our discrete per-step rate to the continuous-time rate of Ref. [29], we reproduce the monitored-Néel entanglement transition (Fig. 8) and identify the volume-to-area crossing $p_c \simeq 0.05$. With $\gamma = p/dt$ and $dt = 0.6$ this gives $\gamma_c \simeq 0.083$, the same order as the reported

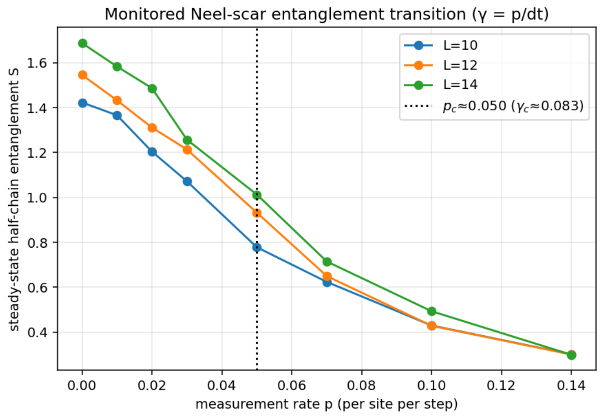

Figure 8: Steady-state half-chain entanglement of the monitored Néel state versus p for $L = 10, 12, 14$; the crossing sets p_c and hence $\gamma_c = p_c/dt$.

$\gamma_c \simeq 0.013 \pm 0.002$; the factor ~ 6 reflects protocol and estimator differences (stroboscopic per-step versus continuous-time monitoring; a crossing estimate versus finite-size scaling) that we have not converted quantitatively.

G Spin-1 Schecter–Iadecola model

The second model is the spin-1 chain [33]

$$H = J \sum_i (S_i^x S_{i+1}^x + S_i^y S_{i+1}^y) + h \sum_i S_i^z + D \sum_i (S_i^z)^2, \tag{3}$$

with J=1, h=1, D=0.1. Its exact scar tower is $|S_n\rangle = (Q^+)^n |\Omega\rangle$, $Q^+ = \sum_j (-1)^j (S_j^+)^2$, $|\Omega\rangle = |m{=}{-1}\rangle^{\otimes L}$. The site operators $\{|m{=}{-1}\rangle, |m{=}{+1}\rangle\}$ realize a pseudospin-$\frac{1}{2}$ (with

$|m{=}0\rangle$ outside), and the tower is the fully symmetric Dicke multiplet of total pseudospin $J = L/2$. With the properly normalized generators $J^+ = Q^+/2$, $J^- = (J^+)^\dagger$, $J^z = \frac{1}{2}\sum_i S_i^z$ one has $[J^z, J^\pm] = \pm J^\pm$, $[J^+, J^-] = 2J^z$, so the Casimir $J^2 = (J^z)^2 + \frac{1}{2}(J^+J^- + J^-J^+)$ equals $J(J{+}1) = \frac{L}{2}(\frac{L}{2}{+}1)$.

We verify numerically that $|S_n\rangle$ are exact eigenstates (residual $\|H\,|S_n\rangle - E_n\,|S_n\rangle\| \sim 10^{-16}$), equally spaced by 2, and that J^2 is constant on the tower with variance $\sim 10^{-13}$ on every scar—versus $\mathrm{Var}\,J^2 \simeq 32$ for PXP scars. Consequently a two-rung scar code has identical J^2 outcome distributions and is an exact decoherence-free subspace of a J^2 measurement, which is why the rescue succeeds here and fails for PXP ($\mathcal{C}_R^{\mathrm{scar}} = 1$, $\Delta\mathcal{C}_R = +0.6$ to $+0.9$ at $L = 6, 8$). Under generic local S^z monitoring, by contrast, neither code is systematically protected: $\Delta\mathcal{C}_R = +0.10, +0.06, -0.02$ at $p = 0.04, 0.08, 0.12$ (ensemble SEM ~ 0.04; 20-pair ensemble, $L = 6$) is scar-favorable at the lowest rate (about 2σ; up to $+0.17$ at $L = 8$) but decays and reverses sign with increasing p, so the scar advantage remains confined to the fine-tuned Casimir probe.

H A single-shot resolvability predictor

The two channels of the main-text criterion combine into one exact, simulation-free scalar: the reference entropy that survives a *single* projective O-measurement of the freshly encoded pair,

$$s_1(O) = \sum_g q_g\, S(\rho_R^{(g)}), \qquad [\rho_R^{(g)}]_{ab} = \frac{\langle b_L|\Pi_g|a_L\rangle}{2q_g}, \tag{4}$$

with Π_g the spectral projectors of O and $q_g = \frac{1}{2}(\langle 0_L|\Pi_g|0_L\rangle + \langle 1_L|\Pi_g|1_L\rangle)$ the Born weights. By construction $s_1 = 1$ iff a single shot leaves the qubit perfectly recoverable—the Knill–Laflamme conditions for the projector set $\{\Pi_g\}$, of which a common-eigenvalue eigenspace (a true DFS) is the simplest realization—and $s_1 = 0$ iff one shot fully reads the qubit out; eigenvalue distinguishability [channel (i)] and non-invariance [channel (ii)] both enter automatically.

Across 65 code–measurement pairs spanning both models and five global operators (Fig. 9), s_1 strongly predicts the monitored $\mathcal{C}_R$ *within each resolving measurement*: Spearman $\rho = +0.85$ (PXP, staggered), $+0.87$ and $+0.81$ (spin-1, Casimir and staggered). Its two failure modes delimit its scope and are themselves instructive: for a non-resolving operator (PXP uniform, $s_1 \approx 1$ for all codes) there is no signal left to predict, and for an operator that does not commute with the Hamiltonian (the PXP Casimir, $[J^2, H] \neq 0$) the dynamics between shots dress the resolvability and shift $\mathcal{C}_R$ systematically below the single-shot estimate. We therefore present s_1 as an operational design quantity—an exact, Monte-Carlo-free screen for candidate code/measurement pairs—rather than a universal collapse variable.

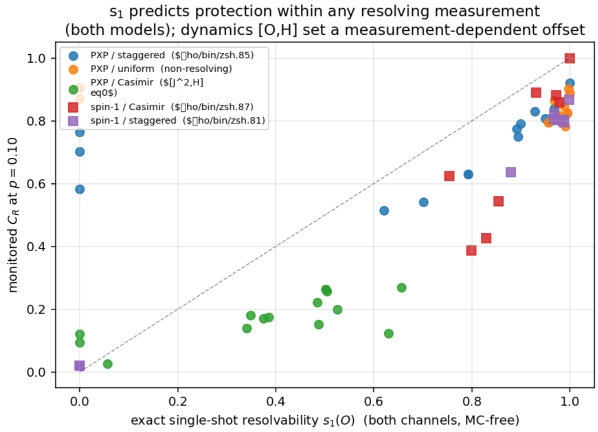

Figure 9: Monitored $\mathcal{C}_R$ ($p = 0.10$) versus the exact single-shot resolvability $s_1(O)$ for 65 code–measurement pairs across both models. Within each resolving measurement the relation is tight; the non-resolving (uniform) and non-commuting (PXP Casimir) families bound the predictor's scope.

I Extensive code under collective and Casimir monitoring

The maximal $k = \lfloor \log_2(L{+}1) \rfloor$-qubit scar code of the main text is compared to a same-dimension thermal code under all three PXP measurement models in Table 1 ($L = 14$, $k = 3$, protected density S_R/k, 8-draw thermal ensemble). The pattern matches the single-qubit analysis: the scar loses under local and (strongly) collective monitoring and is only comparable under the fine-tuned Casimir.

Table 1: Extensive-code density gap $\Delta(S_R/k)$ = scar − thermal ($L = 14$, $k = 3$).

measurement	p=0.04	p=0.08	p=0.12
local S^z	−0.14	−0.06	—
collective M	−0.32	−0.40	−0.43
Casimir J^2	+0.02	+0.03	+0.03

References

[1] Yaodong Li, Xiao Chen, and Matthew P. A. Fisher. "Quantum Zeno effect and the many-body entanglement transition". Phys. Rev. B **98**, 205136 (2018).

[2] Yaodong Li, Xiao Chen, and Matthew P. A. Fisher. "Measurement-driven entanglement transition in hybrid quantum circuits". Phys. Rev. B **100**, 134306 (2019).

[3] Brian Skinner, Jonathan Ruhman, and Adam Nahum. "Measurement-induced phase transitions in the dynamics of entanglement". Phys. Rev. X **9**, 031009 (2019).

[4] Amos Chan, Rahul M. Nandkishore, Michael Pretko, and Graeme Smith. "Unitary-projective entanglement dynamics". Phys. Rev. B **99**, 224307 (2019).

[5] Yimu Bao, Soonwon Choi, and Ehud Altman. "Theory of the phase transition in random unitary circuits with measurements". Phys. Rev. B **101**, 104301 (2020).

[6] Chao-Ming Jian, Yi-Zhuang You, Romain Vasseur, and Andreas W. W. Ludwig. "Measurement-induced criticality in random quantum circuits". Phys. Rev. B **101**, 104302 (2020).

[7] Matthew P. A. Fisher, Vedika Khemani, Adam Nahum, and Sagar Vijay. "Random quantum circuits". Annu. Rev. Condens. Matter Phys. **14**, 335 (2023).

[8] Soonwon Choi, Yimu Bao, Xiao-Liang Qi, and Ehud Altman. "Quantum error correction in scrambling dynamics and measurement-induced phase transition". Phys. Rev. Lett. **125**, 030505 (2020).

[9] Michael J. Gullans and David A. Huse. "Dynamical purification phase transition induced by quantum measurements". Phys. Rev. X **10**, 041020 (2020).

[10] Ruihua Fan, Sagar Vijay, Ashvin Vishwanath, and Yi-Zhuang You. "Self-organized error correction in random unitary circuits with measurement". Phys. Rev. B **103**, 174309 (2021).

[11] Michael J. Gullans and David A. Huse. "Scalable probes of measurement-induced criticality". Phys. Rev. Lett. **125**, 070606 (2020).

[12] Dongheng Qian and Jing Wang. "Coherent information phase transition in a noisy quantum circuit". Phys. Rev. B **112**, L180301 (2025).

[13] J. M. Deutsch. "Quantum statistical mechanics in a closed system". Phys. Rev. A **43**, 2046 (1991).

[14] Mark Srednicki. "Chaos and quantum thermalization". Phys. Rev. E **50**, 888 (1994).

[15] Marcos Rigol, Vanja Dunjko, and Maxim Olshanii. "Thermalization and its mechanism for generic isolated quantum systems". Nature **452**, 854 (2008).

[16] Luca D'Alessio, Yariv Kafri, Anatoli Polkovnikov, and Marcos Rigol. "From quantum chaos and eigenstate thermalization to statistical mechanics and thermodynamics". Adv. Phys. **65**, 239 (2016).

[17] Emanuel Knill and Raymond Laflamme. "Theory of quantum error-correcting codes". Phys. Rev. A **55**, 900 (1997).

[18] Fernando G. S. L. Brandão, Elizabeth Crosson, M. Burak Şahinoğlu, and John Bowen. "Quantum error correcting codes in eigenstates of translation-invariant spin chains". Phys. Rev. Lett. **123**, 110502 (2019).

[19] Ning Bao and Newton Cheng. "Eigenstate thermalization hypothesis and approximate quantum error correction". J. High Energy Phys. **2019**, 152 (2019).

[20] Shozab Qasim and Jason Pollack. "Approximate quantum error correction, eigenstate thermalization and the chaos bound" (2025). arXiv:2510.26758.

[21] Hannes Bernien, Sylvain Schwartz, Alexander Keesling, Harry Levine, Ahmed Omran, Hannes Pichler, Soonwon Choi, Alexander S. Zibrov, Manuel Endres, Markus Greiner, Vladan Vuletić, and Mikhail D. Lukin. "Probing many-body dynamics on a 51-atom quantum simulator". Nature **551**, 579 (2017).

[22] C. J. Turner, A. A. Michailidis, D. A. Abanin, M. Serbyn, and Z. Papić. "Weak er-

godicity breaking from quantum many-body scars". Nat. Phys. **14**, 745 (2018).

[23] C. J. Turner, A. A. Michailidis, D. A. Abanin, M. Serbyn, and Z. Papić. "Quantum scarred eigenstates in a Rydberg atom chain: Entanglement, breakdown of thermalization, and stability to perturbations". Phys. Rev. B **98**, 155134 (2018).

[24] Maksym Serbyn, Dmitry A. Abanin, and Zlatko Papić. "Quantum many-body scars and weak breaking of ergodicity". Nat. Phys. **17**, 675 (2021).

[25] Sanjay Moudgalya, B. Andrei Bernevig, and Nicolas Regnault. "Quantum many-body scars and Hilbert space fragmentation: a review of exact results". Rep. Prog. Phys. **85**, 086501 (2022).

[26] Paolo Zanardi and Mario Rasetti. "Noiseless quantum codes". Phys. Rev. Lett. **79**, 3306 (1997).

[27] D. A. Lidar, I. L. Chuang, and K. B. Whaley. "Decoherence-free subspaces for quantum computation". Phys. Rev. Lett. **81**, 2594 (1998).

[28] He-Ran Wang, Dong Yuan, Shun-Yao Zhang, Zhong Wang, Dong-Ling Deng, and L.-M. Duan. "Embedding quantum many-body scars into decoherence-free subspaces". Phys. Rev. Lett. **132**, 150401 (2024).

[29] Alessio Paviglianiti and Alessandro Silva. "Enhancing revivals via projective measurements in a quantum scarred system". Phys. Rev. Lett. **135**, 090402 (2025).

[30] Cheng-Ju Lin and Olexei I. Motrunich. "Exact quantum many-body scar states in the Rydberg-blockaded atom chain". Phys. Rev. Lett. **122**, 173401 (2019).

[31] Wen Wei Ho, Soonwon Choi, Hannes Pichler, and Mikhail D. Lukin. "Periodic orbits, entanglement, and quantum many-body scars in constrained models: Matrix product state approach". Phys. Rev. Lett. **122**, 040603 (2019).

[32] Soonwon Choi, Christopher J. Turner, Hannes Pichler, Wen Wei Ho, Alexios A. Michailidis, Zlatko Papić, Maksym Serbyn, Mikhail D. Lukin, and Dmitry A. Abanin. "Emergent SU(2) dynamics and perfect quantum many-body scars". Phys. Rev. Lett. **122**, 220603 (2019).

[33] Michael Schecter and Thomas Iadecola. "Weak ergodicity breaking and quantum many-body scars in spin-1 XY magnets". Phys. Rev. Lett. **123**, 147201 (2019).

[34] Sanjay Moudgalya, Stephan Rachel, B. Andrei Bernevig, and Nicolas Regnault. "Exact excited states of nonintegrable models". Phys. Rev. B **98**, 235155 (2018).